BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: BN-IX-XXIII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-A)

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-IX-XXIII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Salad, dirt, sand and water are the types of:

(A) Heterogeneous mixture
(B) Compounds
(C) Elements
(D) Homogeneous mixture

Q2. Which one of the following is considered as molecular ion?

(A) CH^{+1}
(B) CH^{+}_{3}
(C) C_2H_5OH
(D) Cl_2

Q3. Which metal was used as radioactive source in Ruther's Ford experiment?

(A) Uranium metal
(B) Polonium metal
(C) Radium metal
(D) Thorium metal

Q4. Iron (Fe) shows +2 and +3 valances in different compounds, in which block it is placed.

(A) s-block
(B) p-block
(C) d-block
(D) f-block

Q5. Which compound shows electronegativity difference greater than 1.7:

(A) NaCl
(B) HCl
(C) CO_2
(D) NH_3

Q6. Two fluorine atoms share one electron each in their outermost shell to attain electronic configuration of:

(A) Xe
(B) Ar
(C) Kr
(D) Ne

Q7. The pressure that is exerted by gasses due to their:

(A) Circular motion
(B) Linear motion
(C) Random motion
(D) Vibratory motion

Q8. How many grams of solute is required to produce 1.00dm^3 of 3.50 M solution of HCl?

(A) 127.75 g
(B) 128.75 g
(C) 129.75 g
(D) 130.75 g

Q9. If a spoon full of sugar is mixed in a glass of water, what is the water called?

(A) Solute
(B) Solution
(C) Solvent
(D) Suspension

Q10. Which one of the following is not an oxidizing agent?

(A) Potassium iodide
(B) Potassium Manganate
(C) Potassium dichromate
(D) Bromine solutions

Q11. Which of the following compound shows maximum oxidation number of Nitrogen?

(A) NO
(B) NO_2
(C) NO_3
(D) HNO_3

Q12. The elements having high value of ionization energies is called:

(A) Metals
(B) Non-metals
(C) Metalloids
(D) Electropositive elements

SECTION – B

Time Allowed: 2:40Hrs | **Maximum Marks:** 32 | **Paper Code:** BN-IX-XXIII-I

> Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. How many moles are present in 196g of H_2SO_4? (At.mass H = 1amu, S = 32amu, O = 16amu)

ii. Helium follow duplet rule while Argon follow octet rule. Give suitable reason to support your answer.

iii. Differentiate between alkali metals and halogens.

iv. How many elements follow the Duplet rule? Support your answer by their electronic configuration?

v. Classify the following molecules as having single covalent bond and double covalent bond? CO_2, H_2O, O_2, CO

vi. 3.0L of container filled with a gas at a temperature of 25°C. The gas exerts a pressure at 1 atm on the container. If pressure is kept constant, what is the final volume of a gas, if temperature of container is increase to 200°C

vii. Identity solute and solvent in the following:

a) Alcohol in water

b) Mercury amalgam

viii. How can we prepare 500ml of 0.02 molar solution of H_2SO_4 from 2 molar stock solutions?

ix. Write the construction of electrode in Nelson cell also write the products produced at both electrodes.

x. Explain the electroplating on the basis of electrodic reaction.

xi. Give the reason why platinum, gold and silver are used to make jewelry?

SECTION – C

Maximum Marks: 21 | **Paper Code:** BN-IX-XXIII-I

Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) [3] Find out the formula mass of $NaNO_3$ in a.m.u and express it into grams?

b) [4] Draw atomic structure of: $^{234}_{92}U$, $^{235}_{92}U$, $^{236}_{92}U$, $^{12}_{6}C$ and $^{2}_{1}H$.

Q4.

a) [3] What is the difference between alkali metals and alkaline earth metals in terms of their atomic structure? Also write their group number.

b) [4] Explain the reaction that occurs between ammonia and boron trifluoride.

Q5.

a) [3] Find the volume of $2dm^3$ gas if pressure is increased from 300 mm of Hg to 500 mm of Hg.

b) [4] Define supersaturate solution elaborate your answer with suitable example.

Q6.

a) [3] Write any three uses of sodium?

b) [4] Write three electrochemical operations that are used in electrochemical industries.